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(54) **APPARATUS FOR GRINDING SURFACE OF STEEL SHEET**

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CPC **B24B 29/02** (2013.01); **B24B 41/047** (2013.01)

(57)

ABSTRACT

A steel plate surface polishing device comprises a transfer unit transferring a steel plate, a polishing unit positioned over and coupled to the transfer unit to be rotatable around an axis along a direction perpendicular to the transfer unit, polishing a surface the steel plate through a polishing belt, and adjusting a polishing angle of the steel plate according to rotation, and an adhesion unit coupled to one side of the polishing unit to bring the polishing belt into tight contact with the steel plate.

100

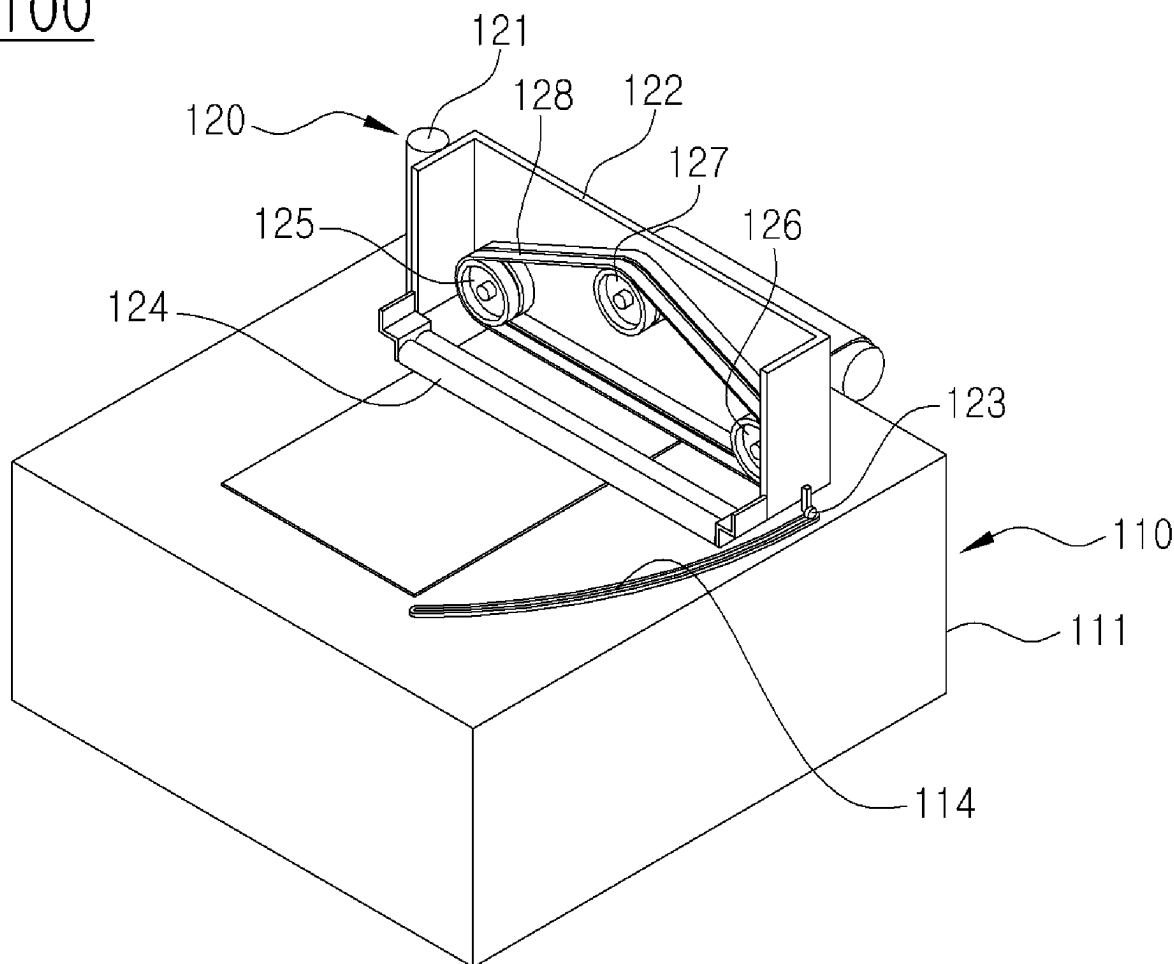


FIG. 2

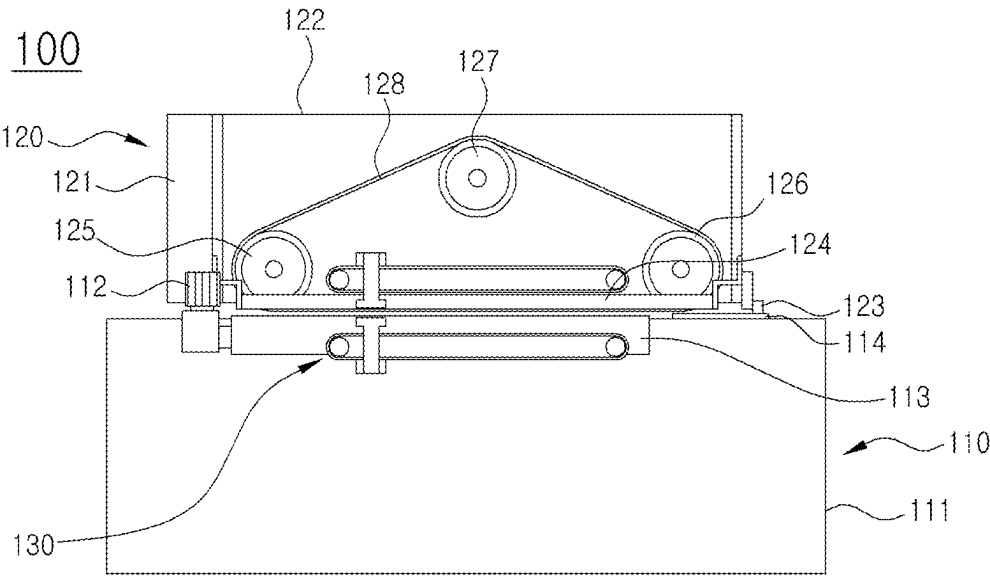
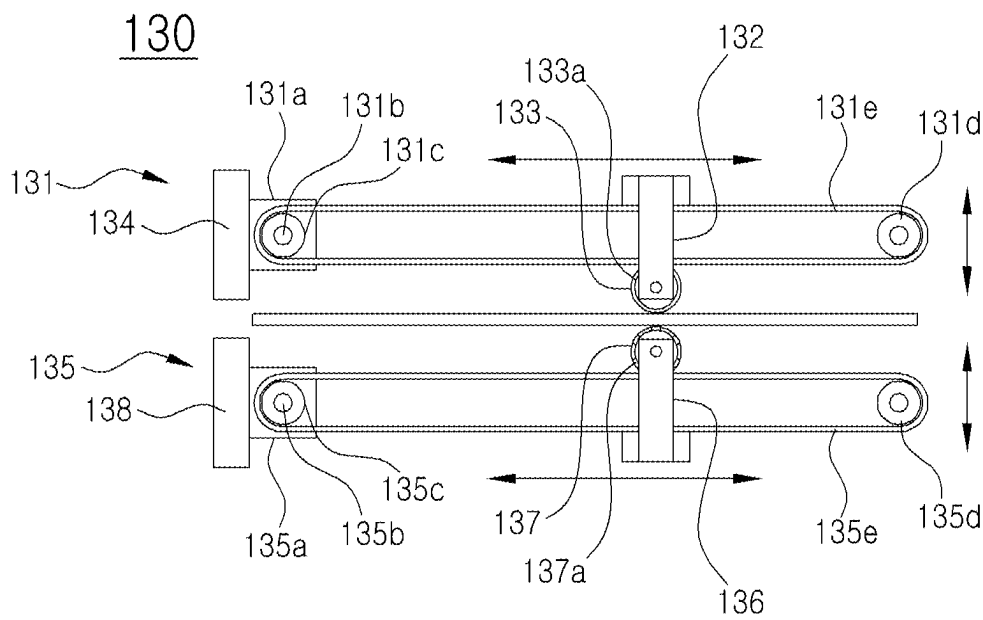


FIG. 3



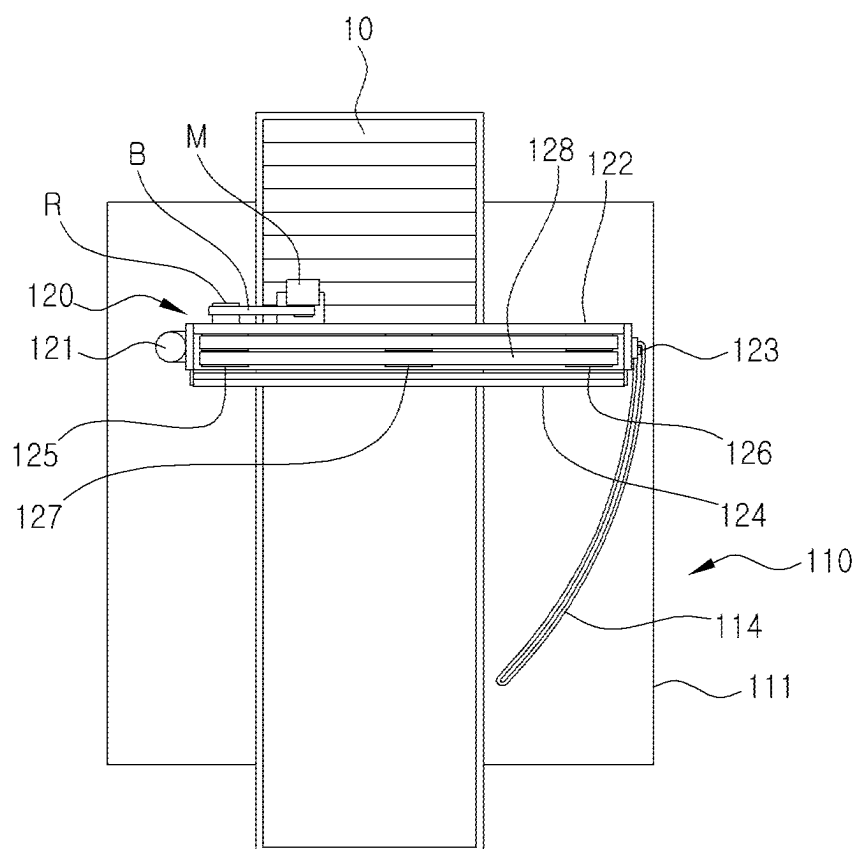


FIG. 5

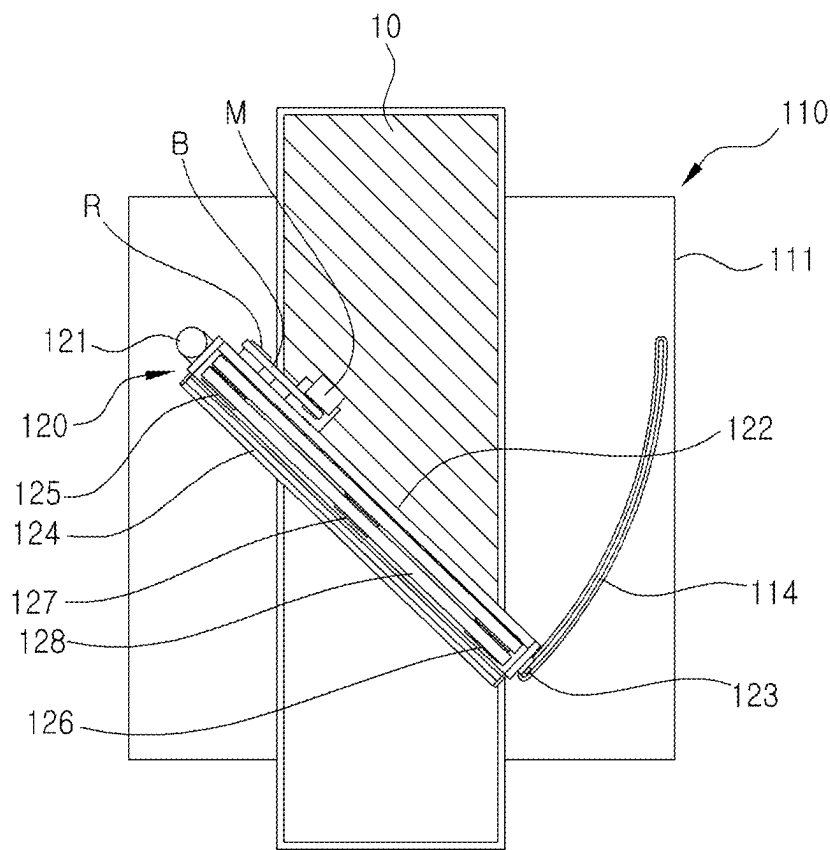


FIG. 6

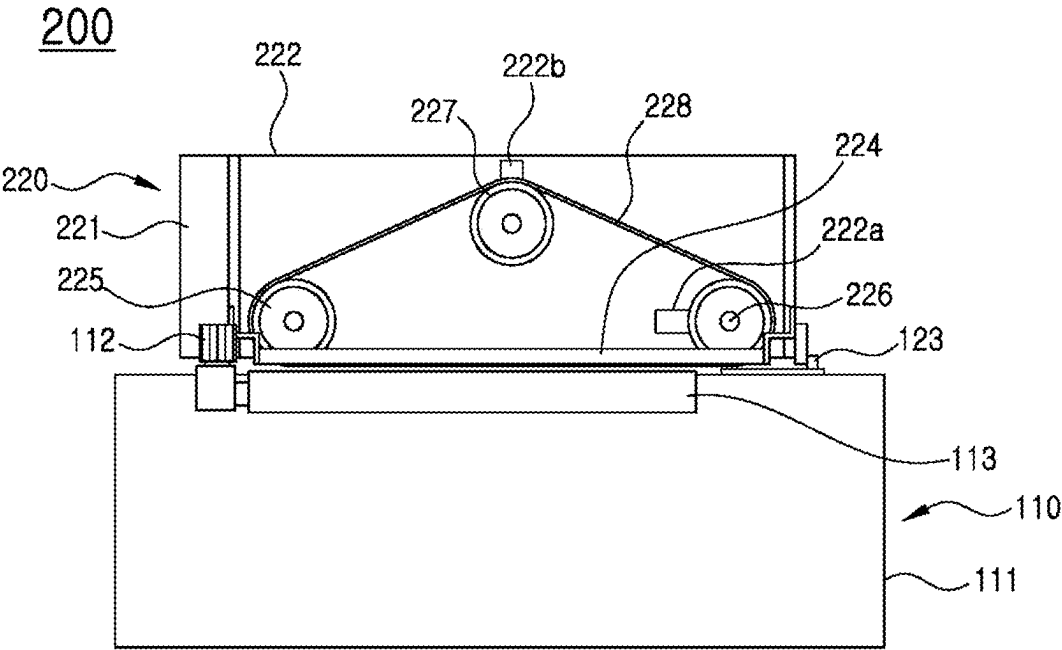


FIG. 7

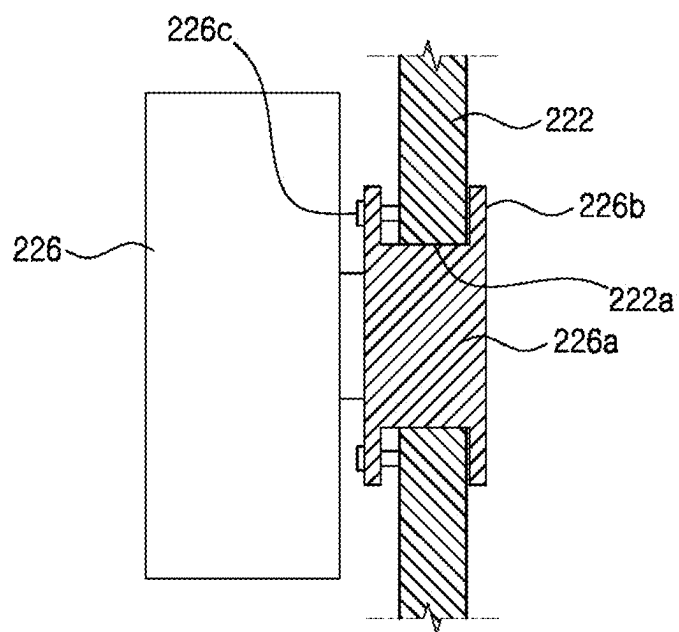


FIG. 8

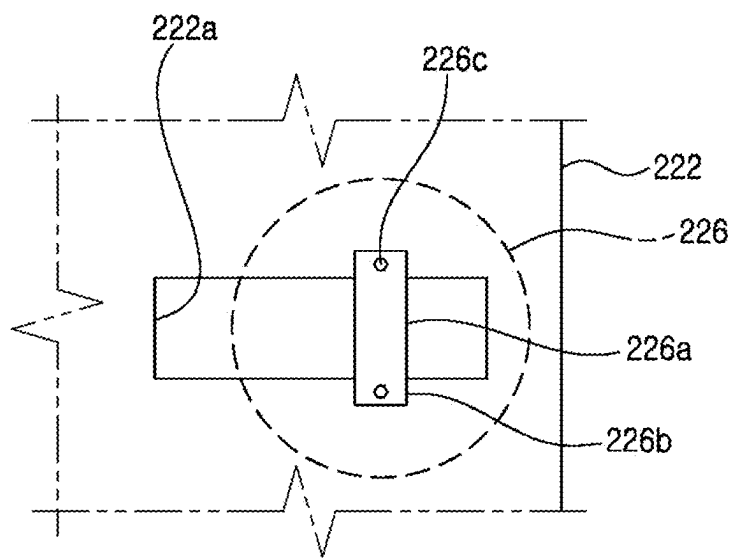


FIG. 9

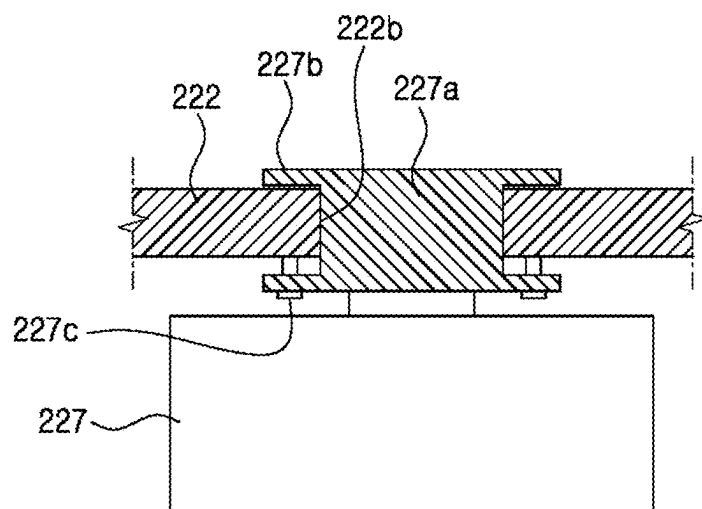


FIG. 10

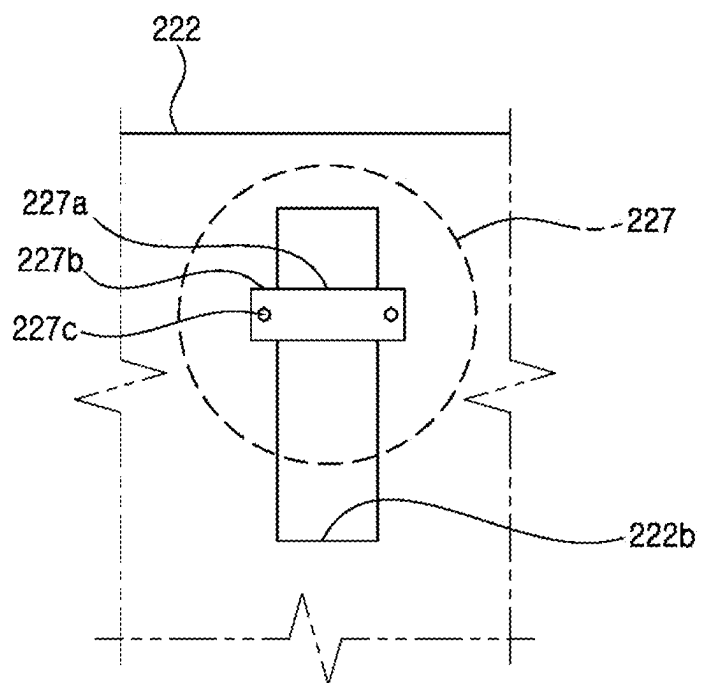
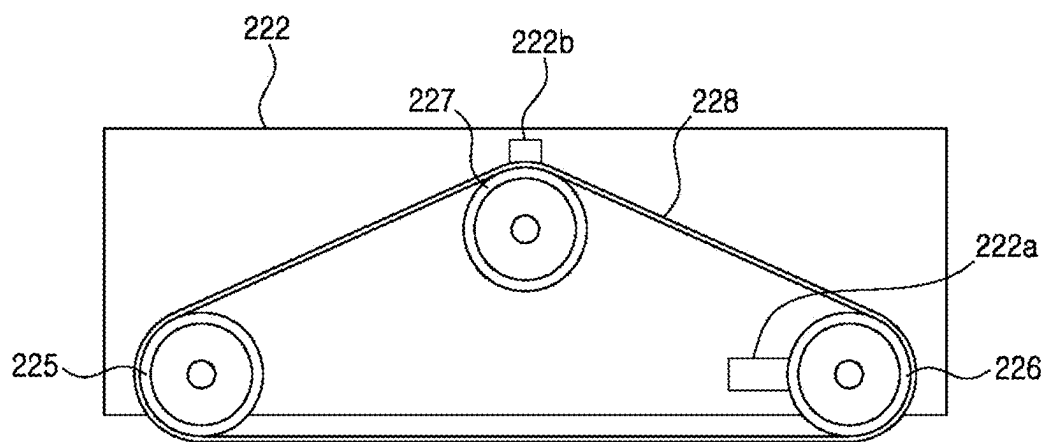
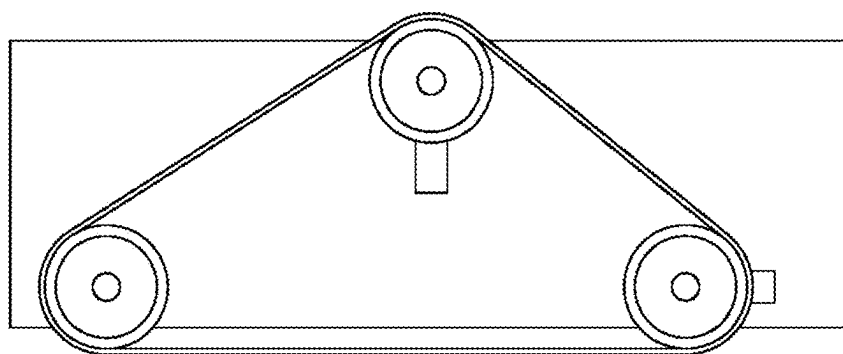


FIG. 11



(a)



(b)

APPARATUS FOR GRINDING SURFACE OF STEEL SHEET

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is a continuation-in-part (CIP) patent application which claims priority to U.S. patent application Ser. No. 17/802,134, filed on Aug. 24, 2022, which is a national-stage application of International Patent Application No. PCT/KR2021/002300, filed on Feb. 24, 2021, which claims priority to Korean Patent Application No. 10-2020-0025444 filed in the Korean Intellectual Property Office on Feb. 28, 2020, the disclosures of which are incorporated by reference herein in their entireties.

TECHNICAL FIELD

[0002] The disclosure relates to a surface polishing device for steel sheets, and more particularly, to a surface polishing device for steel sheets used for polishing the surface of a steel sheet.

DISCUSSION OF THE RELATED ART

[0003] Forming a scratch pattern, like hairlines, on the surface of a metal plate, e.g., a stainless steel plate, may provide a gorgeous decorative effect while maintaining the inherent luster of the metal and may also achieve anti-fingerprint effects. As such, hairline finishing is widely used as a surface processing method for stainless steel or other metal plates.

[0004] In general, conventional devices for forming hairlines on metal plates form hairlines by rotating or reciprocating a sandpaper-wrapped polishing belt or pad on a steel plate continuously uncoiled or a steel sheet cut to a predetermined length along the feeding or conveying direction of the coiled steel plate or sheet to thereby make scratches. The conventional hairline finishing devices form hairlines along the length direction of the coiled steel plate or the steel sheet cut to a predetermined length and have difficulty in forming hairlines along the horizontal direction of the coiled steel plate or steel sheet.

[0005] Therefore, conventionally, products with a hairline pattern in the length direction of the coil or sheet are mainly supplied, and there is no compact, efficient device for automatically forming a hairline pattern in the horizontal direction of the coiled steel plate or steel sheet. In particular, there is no known efficient, compact device capable of forming hairlines in

[0006] the horizontal direction for both metal plates continuously uncoiled from coiled steel plates and steel sheets cut to a predetermined length. Therefore, conventional hairline-finished products mostly have hairlines in the length direction but there are no products with hairlines in the horizontal direction which may give a novel, aesthetic feeling.

[0007] The conventional hairline forming device typically includes one device unit and one polishing belt or polishing pad. Thus, to form mixed hairlines, which include multiple types of hairlines with different thicknesses, on a single metal plate, multiple hairline forming devices are required, which may cause a complicated device configuration and an increase in space occupation.

SUMMARY

[0008] The disclosure has been devised to address the foregoing issues and aims to polish to have various patterns through a minimal headline device while reducing the size and costs of the device. To achieve the foregoing objectives, according to a preferred embodiment of the disclosure,

[0009] a steel plate surface polishing device comprises a transfer unit transferring a steel plate, a polishing unit positioned over and coupled to the transfer unit to be rotatable around an axis along a direction perpendicular to the transfer unit, polishing a surface the steel plate through a polishing belt, and adjusting a polishing angle of the steel plate according to rotation, and an adhesion unit coupled to one side of the polishing unit to bring the polishing belt into tight contact with the steel plate.

[0010] Further, a guide rail is coupled to an upper end of the transfer unit to guide a moving roller coupled to an end of the polishing unit and to fix a rotation angle of the polishing unit.

[0011] Further, the guide rail guides the polishing unit to be rotated at an angle ranging from 0 degrees to 45 degrees from a direction perpendicular to a transfer direction of the transfer unit.

[0012] Further, a plurality of the polishing unit are disposed to be symmetrical with each other in a front/rear direction.

[0013] Further, a driving roller driven by a driving motor is coupled to one side of the polishing unit. A horizontal roller positioned in line with the driving roller to be movable in a transverse direction is coupled. A vertical roller is coupled over and between the driving roller and the horizontal roller to be movable in a longitudinal direction. A polishing belt is coupled integrally to the driving roller, the horizontal roller, and the vertical roller. A length of a lowermost end of the polishing belt is adjusted as the horizontal roller and the vertical roller are moved in the transverse direction and the longitudinal direction and the length of the lowermost end is adjusted as the polishing unit is rotated.

[0014] According to the steel plate surface polishing device of the disclosure, it is possible to form various patterns on the surface of a steel plate by adjusting the polishing angle through rotation of the polishing unit rotatably coupled to the transfer unit.

[0015] In particular, various patterns may be formed by adjusting the angle of a single polishing unit, so that it is possible to simplify and downsize the equipment to save costs.

[0016] Further, polishing unit symmetrical to each other may be provided to enable polishing in further diversified patterns.

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 is a perspective view illustrating a steel plate surface polishing device according to an embodiment of the disclosure.

[0018] FIG. 2 is a front view of FIG. 1 according to the embodiment of the disclosure.

[0019] FIG. 3 is a front view illustrating an adhesion unit according to an embodiment.

[0020] FIGS. 4 and 5 are views illustrating a polishing unit according to the embodiment of the disclosure.

[0021] FIG. 6 is a front view illustrating a steel plate surface polishing device according to a second embodiment of the disclosure.

[0022] FIG. 7 is a side cross-sectional view illustrating a horizontal roller according to the embodiment of the disclosure.

[0023] FIG. 8 is a front view illustrating the horizontal roller according to the embodiment of the disclosure.

[0024] FIG. 9 is a side cross-sectional view illustrating a vertical roller according to the embodiment of the disclosure.

[0025] FIG. 10 is a front view illustrating the vertical roller according to the embodiment of the disclosure.

[0026] FIG. 11 is a view illustrating operations of the horizontal roller and the vertical roller according to an embodiment of the disclosure.

DETAILED DESCRIPTION

[0027] Advantages and features of the disclosure, and methods for achieving the same may be apparent from the embodiments described below with reference to the accompanying drawings.

[0028] However, the disclosure is not limited to the embodiments disclosed herein, and various changes may be made thereto. The embodiments disclosed herein are provided only to inform one of ordinary skilled in the art of the category of the disclosure. The disclosure is defined only by the appended claims. The same reference numeral denotes the same element throughout the specification.

[0029] Hereinafter, the disclosure is described with reference to the drawings for describing a steel plate surface polishing device according to embodiments of the disclosure.

[0030] FIG. 1 is a perspective view illustrating a steel plate surface polishing device according to an embodiment of the disclosure. FIG. 2 is a front view of FIG. 1 according to an embodiment of the disclosure. FIGS. 3 and 4 are views illustrating a polishing unit according to the embodiment of the disclosure.

[0031] Referring to these drawings, the steel plate surface polishing device according to the present embodiment is able to finish the steel plate surface to have various patterns even with a simple process.

[0032] The steel plate surface polishing device 100 according to an embodiment that may provide such an effect includes a transfer unit 110, a polishing unit 120, and an adhesion unit 130.

[0033] The transfer unit 110 transfers a steel plate 10 and includes a lower body 111 installed on the ground.

[0034] A transfer roller 113 driven by a transfer motor 112 is coupled to an upper end of the lower body 111.

[0035] At least one or more transfer rollers 113 may be disposed along the transfer direction of the steel plate.

[0036] It is preferable that the transfer roller 113 is positioned higher than the top surface of the lower body 111.

[0037] This is to prevent the top surface of the lower body 111 from interfering with the polishing unit 120 when the steel plate 10 transferred by the transfer roller 113 is polished by the polishing unit 120 to be described below.

[0038] A guide rail 114 which is curved from one side, as a reference point, of the transfer roller 113 to the other side is coupled to the top end of the lower body 111.

[0039] In this case, the guide rail 114 is curved by about 45 degrees forward along the direction perpendicular to the transfer direction of the transfer roller 113.

[0040] The polishing unit 120 is installed over the transfer unit 110 to polish the surface of the steel plate 10 transferred by the transfer roller 113. The polishing unit 120 includes a rotation shaft 121 coupled to a reference point of the lower body 111 and an upper body 122 coupled to a side of the rotation shaft 121.

[0041] Specifically, the rotation shaft 121 is rotatably coupled to the reference point of the lower body 111.

[0042] The upper body 122 has one side rotatably and integrally coupled to the rotation shaft 121 and the other side extending to the other side of the transfer roller 113.

[0043] A moving roller 123 guided along the guide rail 114 is coupled to a lower portion of the other end of the upper body 122.

[0044] The upper body 122 is rotated by about 45 degrees from the direction perpendicular to the transfer direction of the transfer roller 113, along the guide rail 114, by the moving roller 123.

[0045] The moving roller 123 has a brake (not shown) and may fix the rotation angle of the upper body 122.

[0046] A driving motor M is coupled to one side of the rear of the upper body 122, and a sub roller R connected with the driving motor M by a driving belt B is coupled to a lower end of the one side of the rear.

[0047] A driving roller 125 connected with the sub roller R to be integrally driven is coupled to a lower end of the one side of the front of the upper body 122.

[0048] A horizontal roller 126 disposed in line with the driving roller 125 is rotatably coupled to a lower end of the other side of the front of the upper body 122.

[0049] A vertical roller 127 disposed over and between the driving roller 125 and the horizontal roller 126, is rotatably coupled to the front of the upper body 122.

[0050] In this case, the driving roller 125, the horizontal roller 126, and the vertical roller 127 are disposed in a substantially triangular shape.

[0051] One polishing belt 128 is coupled to integrally surround the driving roller 125, the horizontal roller 126, and the vertical roller 127.

[0052] The polishing belt 128 is rotated together with the horizontal roller 126 and the vertical roller 127 to polish the steel plate 10 according to the driving of the driving roller 125.

[0053] In this case, a pressing roller 124 is rotatably coupled to the frontmost lower end of the upper body 122 to press the steel plate 10.

[0054] The pressing roller 124 presses and fixes the steel plate 10 to prevent the steel plate 10 from being shaken by the polishing belt 128 while being polished.

[0055] Thus, the polishing unit 120 may form hairlines in a diagonal direction between 0 degrees and 45 degrees, as well as hairlines in the horizontal direction, on the steel plate 10 through rotation.

[0056] Further, the driving roller 125, the horizontal roller 126, and the vertical roller 127 may be formed in duplicate. The polishing belt 128 may also be coupled in duplicate.

[0057] Further, the polishing unit 120 may be installed at the front and rear, respectively, of the transfer unit 110 and be symmetrical to each other to polish the steel plate 10 substantially in an "x" shape when rotated at 45 degrees therebetween.

[0058] The adhesion unit 130 is to bring the polishing belt 128 into close contact with the steel plate 10, and includes a first driver 131 coupled to one side of the upper body 122 and a second driver 135 coupled to one side of the lower body 111.

[0059] The first driver 131 includes a first driving motor 131a coupled to one side of the upper body 122, a first driving shaft 131b driven by the first driving motor 131a, and a first rotating roller 131c coupled to the first driving shaft 131b. The first rotating roller 131c is disposed at a position adjacent to the driving roller 125.

[0060] The first driver 131 includes a first interlocking roller 131d spaced apart from the first rotating roller 131c by a predetermined distance and disposed at a position adjacent to the horizontal roller 126 and a first driving belt 131e coupled to the first rotating roller 131c and the first interlocking roller 131d.

[0061] The first reciprocating member 132 is coupled to the first driving belt 131e so as to be movable together according to driving, and the first reciprocating member 132 extends to be positioned between the driving roller 125 and the horizontal roller 126. A first contact roller 133 is coupled to an end of the first reciprocating member 132 so as to tightly contact the upper surface of the lower portion of the polishing belt between the driving roller 125 and the horizontal roller 126.

[0062] A first tread 133a is formed on the circumference of the first contact roller 133 along the transverse direction, which is the moving direction of the first reciprocating member 132.

[0063] The adhesion unit 130 includes a first height adjuster 134 for adjusting the height of the first driver 131 coupled to the upper body 122.

[0064] The first height adjuster 134 may adjust the height of the first contact roller 133 by adjusting the height of the first driver 131.

[0065] The second driver 131 includes a second driving motor 135a coupled to one side of the lower body 111, a second driving shaft 135b driven by the second driving motor 135a, and a second rotating roller 135c coupled to the second driving shaft 135b. The second rotating roller 135c is disposed at a position facing the first rotating roller 131c.

[0066] The second driver 135 includes a second interlocking roller 135d spaced apart from the second rotating roller 135c by a predetermined distance and disposed at a position facing the first interlocking roller 131d, and a second driving belt 135e coupled to the second rotating roller 135c and the second interlocking roller 135d.

[0067] The second reciprocating member 136 is coupled to the second driving belt 135e so as to be movable together according to driving, and the second reciprocating member 136 extends symmetrically with the first reciprocating member 132.

[0068] A second contact roller 137 is coupled to an end of the second reciprocating member 136 so as to tightly contact the lower surface of the steel plate 10.

[0069] A second tread 137a disposed in the longitudinal direction, which is the direction in which the steel plate 10 is transferred, is formed on the circumference of the second adhesion roller 137. The second tread 137a is disposed to be perpendicular to the first tread 133a.

[0070] The adhesion unit 130 includes a second height adjuster 138 for adjusting the height of the second driver 135, coupled to the lower body 111.

[0071] The second height adjuster 138 may adjust the height of the second contact roller 137 by adjusting the height of the second driver 135.

[0072] Meanwhile, the first driver 131 and the second driver 135 rotate repeatedly in one direction and the other direction so that the first reciprocating member 132 and the second reciprocating member 136 reciprocate between the two opposite sides.

[0073] FIG. 6 is a front view illustrating a steel plate surface polishing device according to an embodiment of the disclosure. FIG. 7 is a side cross-sectional view illustrating a horizontal roller according to the embodiment of the disclosure. FIG. 8 is a front view illustrating the horizontal roller according to an embodiment of the disclosure. FIG. 9 is a side cross-sectional view illustrating a vertical roller according to an embodiment of the disclosure. FIG. 10 is a front view illustrating the vertical roller according to an embodiment of the disclosure. FIG. 11 is a view illustrating operations of the horizontal roller and the vertical roller according to an embodiment of the disclosure.

[0074] The steel plate surface polishing device 200 according to the present embodiment includes a transfer unit 110 and a polishing unit 220.

[0075] The transfer unit 110 transfers a steel plate 10 and includes a lower body 111 installed on the ground.

[0076] A transfer roller 113 driven by a transfer motor 112 is coupled to an upper center of the lower body 111.

[0077] The polishing unit 220 is installed over the transfer unit 110 to polish the surface of the steel plate 10 transferred by the transfer roller 113. The polishing unit 220 includes a rotation shaft 221 coupled to a reference point of the lower body 111 and an upper body 222 coupled to a side of the rotation shaft 221.

[0078] A driving motor M is coupled to one side of the rear of the upper body 222, and a sub roller R connected with the driving motor M by a driving belt B is coupled to a lower end of the one side of the rear.

[0079] A driving roller 225 connected with the sub roller R to be integrally driven is coupled to a lower end of the one side of the front of the upper body 222.

[0080] A horizontal roller 226 disposed in line with the driving roller 225 is coupled to a lower end of the other side of the front of the upper body 222, rotatably and movably in the traverse direction.

[0081] A vertical roller 227 disposed over and between the driving roller 225 and the horizontal roller 226, is coupled to the front of the upper body 222, rotatably and movably in the longitudinal direction.

[0082] In this case, the driving roller 225, the horizontal roller 226, and the vertical roller 227 are disposed in a substantially triangular shape.

[0083] One abrasive belt 228 is coupled to integrally surround the driving roller 225, the horizontal roller 226, and the vertical roller 227.

[0084] The polishing belt 228 is rotated together with the horizontal roller 226 and the vertical roller 227 to polish the steel plate 10 according to the driving of the driving roller 225.

[0085] In this case, a pressing roller 224 is rotatably coupled to the frontmost lower end of the upper body 222 to press the steel plate 10.

[0086] The pressing roller 224 presses and fixes the steel plate 10 to prevent the steel plate 10 from being shaken by the polishing belt 228 while being polished.

[0087] Meanwhile, the horizontal roller 226 and the vertical roller 227 may be movably coupled to the upper body 222 and be fixed in a moved position and, as moved, be able to adjust the lowermost length of the polishing belt 228.

[0088] Specifically, a first long hole 222a is formed, in the transverse direction, in the lower end of the other side of the front of the upper body 222, and a first moving member 226a is movably coupled to the first long hole 222a.

[0089] The horizontal roller 226 is rotatably coupled to the front of the first moving member 226a.

[0090] Further, first guide wings 226b extending in the upper and lower directions of the first long hole 222a are formed in the first moving member 226a in the front/rear direction of the upper body 222.

[0091] Here, a first fixing bolt 226c for fixing the position of the first moving member 226a is fastened to the first guide wing 226b positioned in front of the upper body 222 among the first guide wings 226b.

[0092] In this case, an end of the first fixing bolt 226c presses the front of the upper body 222 to fix the first moving member 226a according to the degree of fastening.

[0093] A second long hole 222b is formed, in the longitudinal direction, in the upper center of the front of the upper body 222, and a second moving member 227a is movably coupled to the second long hole 222b.

[0094] The vertical roller 227 is rotatably coupled to the front of the second moving member 227a.

[0095] Further, second guide wings 227b extending in two lateral directions of the second long hole 222b are formed in the second moving member 227a in the front/rear direction of the upper body 222.

[0096] Here, a second fixing bolt 227c for fixing the position of the second moving member 227a is fastened to the second guide wing 227b positioned in front of the upper body 222 among the second guide wings 227b.

[0097] In this case, an end of the second fixing bolt 227c presses the front of the upper body 222 to fix the second moving member 227a according to the degree of fastening.

[0098] Accordingly, the positions of the horizontal roller 226 and the vertical roller 227 may be adjusted.

[0099] Meanwhile, the length of the lowermost end of the polishing belt 228 is adjusted according to the positions of the horizontal roller 226 and the vertical roller 227.

[0100] Specifically, the length of the lowermost end of the polishing belt 228 is reduced when the horizontal roller 226 moves in one lateral direction and increased when the horizontal roller 226 moves in the opposite lateral direction.

[0101] In this case, the vertical roller 227 is moved upward, when the horizontal roller 226 moves in the one lateral direction, and moved downward when moving in the opposite direction so as to constantly maintain the tension of the polishing belt 228 according to the upper/lower movements.

[0102] Here, since the contact surface of the polishing belt 228 with the steel plate 10 varies depending on the rotation angle of the polishing unit, the length of the lowermost end thereof is preferably adjusted to correspond thereto.

[0103] For example, when the rotation angle of the polishing unit 220 is 45 degrees, the polishing unit 220 is positioned in an oblique direction with respect to the transfer roller 113, so that the area of contact with the transfer roller 113 is larger than that when the polishing unit 220 is at a rotation angle of 0 degrees.

[0104] Therefore, it is preferable to adjust the length of the lowermost end of the polishing belt 228 according to the rotation angle of the polishing unit 220 so that the area in contact with the transfer roller 113 is the same.

[0105] Here, when the polishing unit 220 is positioned in a direction perpendicular to the transfer roller 113, the rotation angle is 0 degrees, and the area of contact with the polishing belt 228 is relatively short.

[0106] Accordingly, the polishing belt 228 provides an optimal polishing area by adjusting the length of the lowermost end through the horizontal roller 226 and the vertical roller 227 according to the rotation angle of the polishing unit 220.

[0107] By such a configuration, the polishing unit 220 may form hairlines in a diagonal direction between 0 degrees and 45 degrees, as well as hairlines in the horizontal direction, on the steel plate 10 according to the rotation.

[0108] According to the steel plate surface polishing device of the disclosure, it is possible to form various patterns on the surface of a steel plate by adjusting the polishing angle through rotation of the polishing unit rotatably coupled to the transfer unit.

[0109] In particular, various patterns may be formed by adjusting the angle of a single polishing unit, so that it is possible to simplify and downsize the equipment to save costs.

[0110] Further, polishing units symmetrical to each other may be provided to enable polishing in further diversified patterns.

[0111] It will be appreciated by one of ordinary skill in the art that the disclosure may be implemented in other various specific forms without changing the essence or technical spirit of the disclosure. Thus, it should be noted that the above-described embodiments are provided as examples and should not be interpreted as limiting. It should be noted that the scope of the disclosure is defined by the appended claims rather than the described description of the embodiments and include all modifications or changes made to the claims or equivalents of the claims.

What is claimed is:

1. A steel plate surface polishing device, comprising:

a transfer unit transferring a steel plate;

a polishing unit positioned over and coupled to the transfer unit to be rotatable around an axis along a direction perpendicular to the transfer unit, polishing a surface the steel plate through a polishing belt, and adjusting a polishing angle of the steel plate according to rotation; and

an adhesion unit coupled to one side of the polishing unit to bring the polishing belt into tight contact with the steel plate.

2. The steel plate surface polishing device of claim 1, wherein a guide rail is coupled to an upper end of the transfer unit to guide a moving roller coupled to an end of the polishing unit and to fix a rotation angle of the polishing unit.

3. The steel plate surface polishing device of claim 2, wherein the guide rail guides the polishing unit to be rotated at an angle ranging from 0 degrees to 45 degrees from a direction perpendicular to a transfer direction of the transfer unit.

4. The steel plate surface polishing device of claim 3, wherein a plurality of the polishing unit are disposed to be symmetrical with each other in a front/rear direction.

5. The steel plate surface polishing device of claim 3, wherein a driving roller driven by a driving motor is coupled to one side of the polishing unit, wherein a horizontal roller positioned in line with the driving roller to be movable in a transverse direction is coupled, wherein a vertical roller is coupled over and between the driving roller and the horizontal roller to be movable in a longitudinal direction, wherein a polishing belt is coupled integrally to the driving roller, the horizontal roller, and the vertical roller, wherein a length of a lowermost end of the polishing belt is adjusted as the horizontal roller and the vertical roller are moved in the transverse direction and the longitudinal direction and the length of the lowermost end is adjusted as the polishing unit is rotated.

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